

LEVEL INSTRUMENTS
LEVEL SWITCHES, GAGE GLASSES,
(EXTERNAL FLOAT BUT NOT DP CELLS)

TYPE	DETAIL (MOUNTED ON EQUIPMENT)	DETAIL (MOUNTED ON STAND PIPE)	P&ID SYMBOL
LEVEL SWITCH			
LEVEL GAGE			
LEVEL CONTROLLER LEVEL TRANSMITTER			

DP INSTRUMENT

TYPE	DETAIL (MOUNTED ON EQUIPMENT)	DETAIL (MOUNTED ON STAND PIPE)	P&ID SYMBOL
LEVEL LOW PRESSURE			
FLOW LOW PRESSURE			
LEVEL (DIAPHRAGM SEAL)			
FLOW (ABOVE 900#)			

PRESSURE INSTRUMENTS

TYPE	DETAIL (MOUNTED ON EQUIPMENT)	DETAIL (MOUNTED ON PIPE)	P&ID SYMBOL
SINGLE PRESSURE INSTRUMENT			
MULTIPLE PRESSURE INSTRUMENT			
DIAPHRAGM SEAL TYPE PRESSURE GAGE			
SINGLE PRESSURE INSTRUMENT (ABOVE 900 #)			
MULTIPLE PRESSURE INSTRUMENT (ABOVE 900 #)			
DIAPHRAGM SEAL TYPE PRESSURE GAGE (ABOVE 900 #)			

TEMPERATURE INSTRUMENTS (TE/TW/TG)

TYPE	DETAIL (MOUNTED ON EQUIPMENT)	DETAIL (MOUNTED ON PIPE)	P&ID SYMBOL
TEMPERATURE GAGE			
TEMPERATURE GAGE (600 # & LOWER)			
TEMPERATURE GAGE (600 # & LOWER)			
GAUGEBOARD TYPE LEVEL INDICATOR			

NOTES:

- FOR TYPICAL BRIDLE (MOUNTED) DP CELL, DISPLACEMENT LT, LG & LS, PLEASE SEE DWG.XB11A-0000-07 P&ID SYMBOL IS THE SAME AS THIS DWG.
- INSTRUMENT CONNECTION ON JACKETED LINE SHALL BE REFERRED TO EACH P&ID.

Att-7

A

B

C

D

E

F

A

B

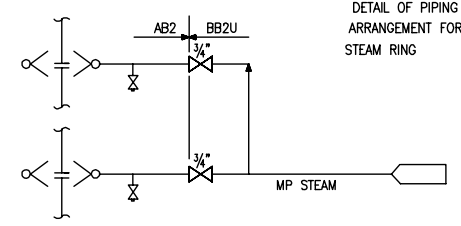
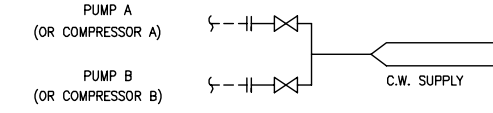
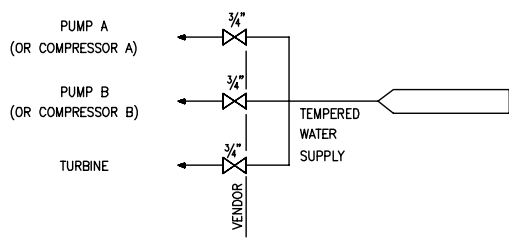
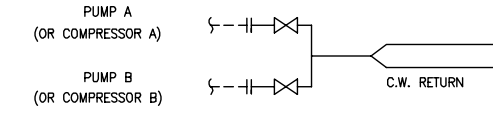
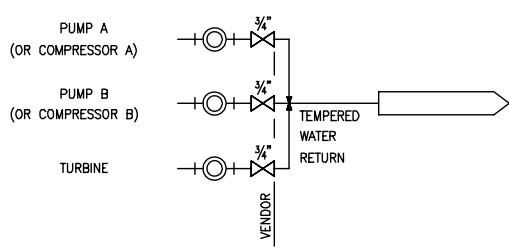
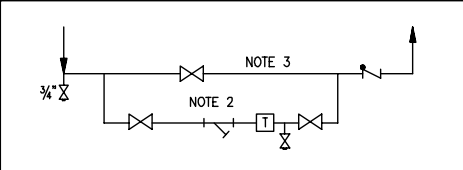
C

D

E

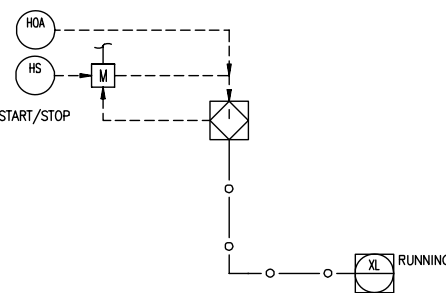
F

TYPICAL DRAWING

P&ID SYMBOL	DETAIL DRAWING	P&ID SYMBOL	DETAIL DRAWING
 NOTE 1	 DETAIL OF PIPING ARRANGEMENT FOR STEAM RING	 SW	 PUMP A (OR COMPRESSOR A) PUMP B (OR COMPRESSOR B) C.W. SUPPLY
 (C-) (P-) (CT-) (PT-)	 PUMP A (OR COMPRESSOR A) PUMP B (OR COMPRESSOR B) TURBINE VENDOR TEMPERED WATER SUPPLY	 RW	 PUMP A (OR COMPRESSOR A) PUMP B (OR COMPRESSOR B) C.W. RETURN
 (C-) (P-) (CT-) (PT-)	 PUMP A (OR COMPRESSOR A) PUMP B (OR COMPRESSOR B) TURBINE VENDOR TEMPERED WATER RETURN	STEAM TRAP DETAIL DRAWING  NOTE 2 NOTE 3 NOTE 3 P&ID SYMBOL  T	

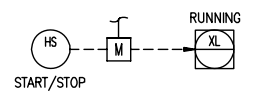
MOTOR AUTO START/STOP DETAIL "A"

(NOTE 4)



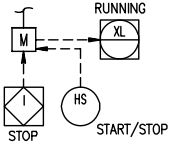
MOTOR MANUAL START/STOP DETAIL 'B'

(NOTE 4)



MOTOR MANUAL START AUTO STOP DETAIL "C"

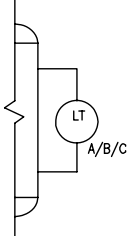
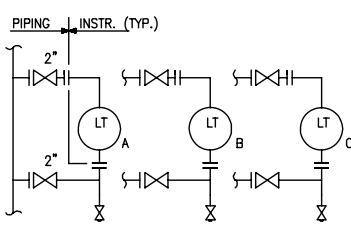
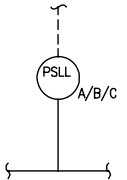
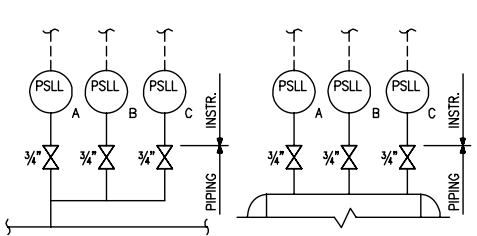
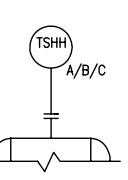
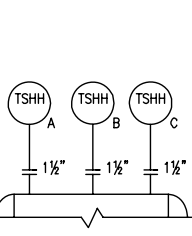
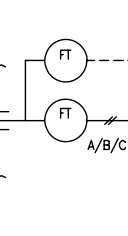
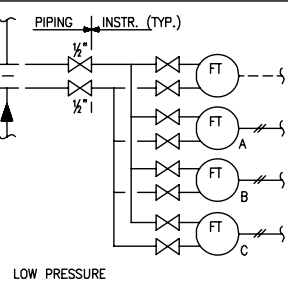
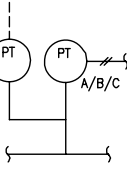
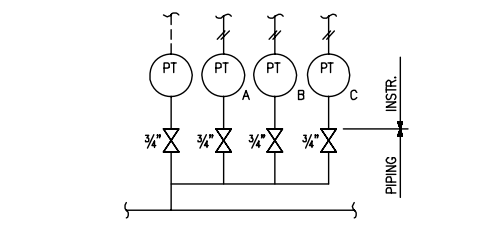
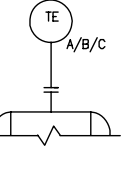
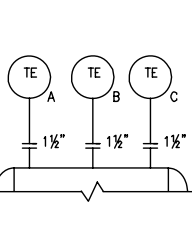
(NOTE 4)



NOTES:

1. VALVES SHOULD BE AT GRADE.
2. TRAP WITH INTEGRAL STRAINER. THE Y-STRAINER CAN BE DELETED.
3. THE BYPASS LINE AND VALVE CAN BE DELETED FOR LOW POINT TRAP SETS.
4. ALL INSTRUMENT TAG NO. (STATUS & SWITCHES) WILL BE THE SAME AS MOTOR (EQUIPMENT) TAG NO. THE DETAILS WILL BE APPLICABLE TO ALL ISBL & OSBL P&IDS EXCEPT FOR FIRE WATER SYSTEM. THE ADDITIONAL FUNCTIONS SHOULD BE INDICATED ON RELEVANT P&ID.

(2/3 LOGIC PIPING)

P&ID SYMBOL	DETAIL DRAWING	P&ID SYMBOL	DETAIL DRAWING	P&ID SYMBOL	DETAIL DRAWING	P&ID SYMBOL	DETAIL DRAWING
 LT A/B/C	 PIPING 2" INSTR. (TYP.)	 PSLL A/B/C	 PSLL A PSLL B PSLL C 3/4" 3/4" 3/4" INSTR. (TYP.)	 TSHH A/B/C	 TSHH A TSHH B TSHH C 1 1/2" 1 1/2" 1 1/2"	 FT A/B/C	 PIPING 1/2" INSTR. (TYP.) LOW PRESSURE
		 PT A/B/C	 PT A PT B PT C 3/4" 3/4" 3/4" INSTR. (TYP.)	 TE A/B/C	 TE A TE B TE C 1 1/2" 1 1/2" 1 1/2"		